

# Stockrium: A Tool for Financial Strategy Experimentation

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**Abstract.** Vietnam’s stock market is among the most retail-driven in the region, yet ordinary investors must reconcile technical, fundamental and news signals across fragmented, largely English-centric tools. We present *Stockrium*, an agentic decision-support system for Vietnamese equities that unifies these signals with recommendation generation. Its core is a multi-agent LangGraph pipeline: an orchestrator dispatches, in parallel, technical, fundamental and news-sentiment agents whose outputs an aggregator fuses into a buy/wait recommendation with horizon-dependent risk targets, following a single principle in which the language model analyzes while deterministic code decides, so that the confidence score, verdict and risk bounds are reproducible and explainable. The pipeline is configurable: users enable or disable each agent and choose the indicators and fundamental metric groups that drive it, making it a flexible testbed for their own strategies rather than a rigid recommender. Backtesting on VN30<sup>1</sup> over a three-year window shows the system behaves as designed: far more selective than the unfiltered technical engine, cutting average drawdown by about a third and almost never trading into downtrends, while matching the engine’s median return at a comparable Sharpe ratio with markedly less risk. As the window is predominantly bullish and lightly traded, we report disciplined, risk-aware behaviour rather than a statistically established return edge. The system is decision-support only and confined to Vietnamese equities.

**Keywords:** Agentic AI · Large language models · Multi-agent systems · Investment decision support · Vietnamese stock market

## 1 Introduction

Vietnam’s stock market has a large retail base: by the end of 2025 domestic investors held about 11.9 million securities accounts (some 11.6% of the population) and around 90% of daily trading value [1,2], ahead of FTSE Russell’s

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<sup>1</sup> VN30: the index of the 30 largest and most-liquid stocks on the Ho Chi Minh Stock Exchange (HOSE).

coming upgrade to Secondary Emerging [3]. Yet the market is led by experienced foreign and proprietary desks toward whom retail investors herd [5]; the non-professionals forming most of the volume lack experience, are prone to losses [4], and follow KOLs<sup>2</sup> in online groups [6] with no way to backtest a strategy of their own.

Existing tools do little to help: consumer apps such as SSI give generic, black-box tips, professional platforms such as TradingView cover only technical analysis, and across all of them the technical, fundamental and news lenses stay fragmented in largely English-centric tools. What is needed is a tool that reconciles these sources in Vietnamese and returns a clear, explainable recommendation a user can trust, adapt and test. A naive LLM cannot fill the gap—it hallucinates and is hard to reproduce or explain when real money is at stake—so a hybrid design is needed, confining the model to reading and summarizing while explicit, auditable rules make every decision.

We present *Stockrium*, a multi-agent LangGraph pipeline for Vietnamese equities: an orchestrator dispatches, in parallel, a technical, a fundamental and a news-sentiment agent whose outputs an aggregator fuses into a buy/wait recommendation with risk-managed targets, under one principle: *the large language model analyzes the evidence and decides the action, while deterministic rules keep it within safe risk bounds*. It is configurable rather than rigid: the user toggles agents and picks the indicators and fundamental groups that feed the analysis, serving amateurs and semi-professionals alike as a unified recommender across the three lenses and a chart-based backtesting testbed for their own strategies.

This paper makes the following contributions:

- A multi-agent LangGraph framework unifying technical, fundamental and news analysis with recommendation for Vietnamese equities, with three analysis agents running in parallel over a shared state and an aggregator producing the final buy/wait decision.
- A fusion-and-decision mechanism: confidence renormalized over active sources, plus a hybrid LLM/rule decision with horizon-dependent risk bounds and a minimum reward-to-risk constraint, keeping verdicts explainable and reproducible.
- A configurable design letting users toggle agents and select indicators and fundamental metric groups, a flexible testbed for their own strategies.
- A purpose-built Vietnamese backtesting tool that overlays trades on the price chart and evaluates strategies with significance testing, extensible to more advanced AI algorithms.
- A curated, research-ready dataset for the Vietnamese market covering all three lenses—price and volume history, company fundamentals, and news with sentiment labels—which we plan to release upon publication.

*Scope.* The system is *decision-support* only: it produces analysis and recommendations but does *not* execute orders, connect to a brokerage, or manage portfolio.

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<sup>2</sup> Key Opinion Leaders

lios, and its output is not financial advice. Its coverage is confined to Vietnamese equities, from which all data, tickers and indices are drawn.

## 2 Related Work

Stockrium builds on four strands of prior work.

*Multi-agent LLM systems for trading.* The closest is TradingAgents [13], which casts a trading firm as a society of LLM agents—analysts, bull/bear researchers, a trader and a risk team—that communicate through structured outputs and natural-language debate, reporting gains on US equities over single-agent and rule baselines. Its role specialization and structured communication inform our design, but it runs on US-centric data and lets the agents debate their way to the decision, which is costly and sensitive to prompt and sampling variation. FinMem [14] instead gives a single LLM trader a layered memory; like TradingAgents, it keeps the decision inside the model. Stockrium adapts the multi-agent idea to the Vietnamese market but moves every decision out of the LLM into deterministic rules and adds per-agent and per-indicator configuration.

*Finance-specific LLMs.* BloombergGPT [11] and the open-source FinGPT [12] improve financial NLP such as sentiment classification and summarization, but they are analytical *assistants*: they neither fuse multiple lenses nor produce risk-managed, backtested recommendations, and none targets Vietnamese equities. We therefore use a strong general model for the language tasks and build the decision logic around it.

*Financial news sentiment analysis.* FinBERT and its successors [8] and FinBERT-LSTM models [7] turn financial text into a sentiment signal that aids price prediction, but they are opaque one-step predictors, not an explainable recommendation that reconciles sentiment with technical and fundamental evidence—the role of our news agent.

*The Vietnamese market.* PhoBERT-based classifiers label stock-article sentiment well [9], news sentiment measurably moves returns [10], and behavioural work documents herding and FOMO among newer investors [5,6]; for price forecasting, LSTM, BiLSTM and Transformer models and conventional machine learning have been applied to the VN-Index [15,16,17]. This work is largely single-source and predictive, rarely LLM-based or risk-calibrated, and offers no interactive tool for fusing signals and backtesting.

To our knowledge no prior system simultaneously (i) unifies technical, fundamental and news analysis with recommendation for the Vietnamese market, (ii) uses a hybrid LLM/rule design that stays explainable and reproducible, (iii) enforces horizon-dependent risk constraints, and (iv) provides a configurable, chart-based backtesting testbed. This paper addresses that gap.

## 3 System Architecture

Stockrium follows a client-server architecture (Fig. 2). On the client side, a mobile application and a web application serve complementary roles. The mobile

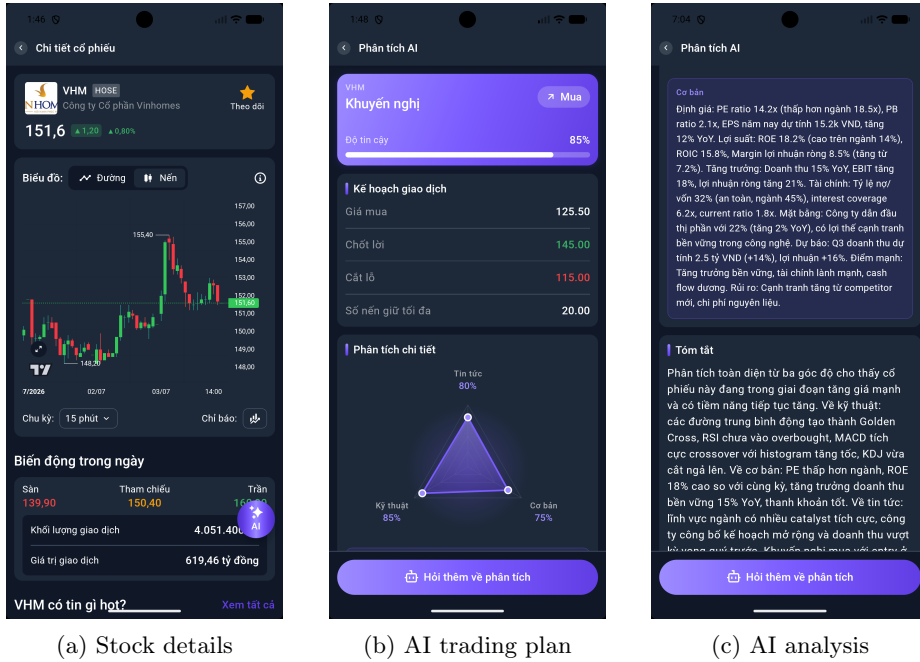


Fig. 1. Stockrium interface captures

application focuses on the core, everyday functions: displaying market information and delivering per-stock analysis and recommendations to the user. The web application focuses on backtesting and on analysis across one or several tickers, letting users configure an analysis (choosing which agents, technical indicators and fundamental metric groups to use), request recommendations, and inspect backtests with the simulated trades drawn on the price chart. Both clients are built with React Native and Expo, sharing most of their code across mobile and web. On the server side, a core service, implemented with FastAPI in Python, exposes the application’s main functions through an API and coordinates three components: a database, a managed PostgreSQL instance hosted on Supabase, that stores the market, fundamental and news data; the multi-agent layer (Section 4), built with LangGraph, that performs the analysis; and a connection to an external LLM provider used by the agents. Keeping a thin client separate from a server that owns the data, the agents and the model access places the heavy analysis and the model credentials on the server, while the clients stay lightweight and interchangeable.

## 4 The Multi-Agent Framework

We frame Vietnamese stock decision support as a multi-source fusion problem: given a ticker and an investor risk profile, the system fuses technical, fundamental

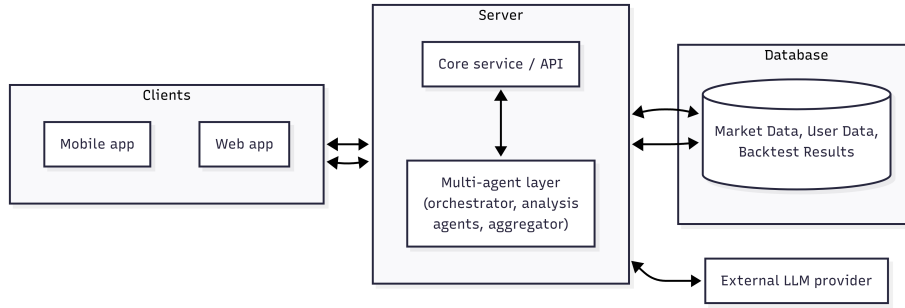


Fig. 2. Client-server architecture

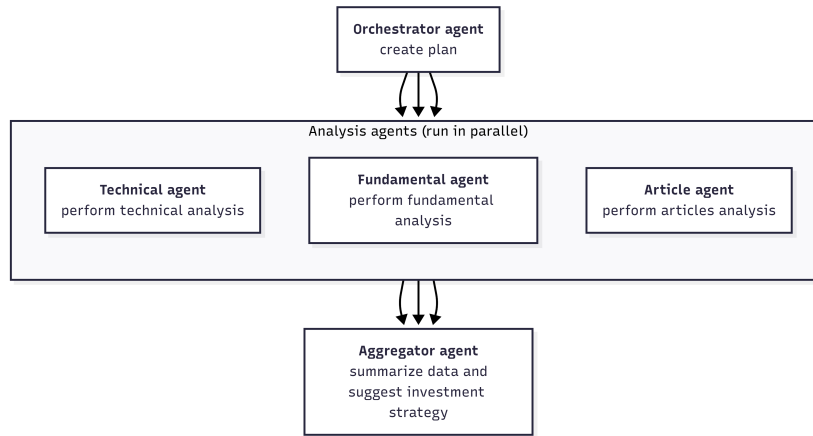


Fig. 3. The multi-agent analysis pipeline

and news-sentiment evidence into a structured, actionable recommendation. It is realized as a LangGraph `StateGraph` of five agents sharing one typed state (Fig. 3): an *orchestrator* that plans the analysis, three *analysis agents* (technical, fundamental and news) that run in parallel, and an *aggregator* that fuses their outputs and decides. Throughout, the language model analyzes the evidence and decides the action while deterministic rules keep it within safe bounds: indicator scores, fusion confidence and risk bounds are computed by explicit, auditable rules that also validate the verdict, keeping outputs reproducible and explainable. The pipeline is also *configurable*: through a `data_selection` object the user enables or disables each agent and picks which technical indicators and fundamental metric groups feed the analysis, so it serves as a testbed for user-defined strategies. We describe each agent below, then how they coordinate (Section 4.6).

### 4.1 Orchestrator Agent

The orchestrator translates a free-form request and the investor’s risk profile into a machine-readable *plan* for the downstream agents, using structured decoding (GPT-4o at temperature 0) so the plan is always a valid Pydantic object. The risk profile’s `period` (`short_term`/`mid_term`/`long_term`, defaulting to `short_term`) is mapped to a candle interval and look-back window: `short_term` to hourly candles over 180 days, `mid_term` to 4-hour candles over 180–365 days, and `long_term` to daily candles over  $\geq 730$  days, so each horizon is analyzed at a granularity matching the intended holding period (the database stores minute-level candles alongside daily ones and aggregates them on demand). The orchestrator never invents a ticker absent from the input, always emits parameters for all three agents, and in `manual` mode forwards a user-supplied plan unchanged.

### 4.2 Technical Analysis Agent

The technical agent computes five standard indicators on the planned candles, entirely in NumPy/Pandas so their smoothing conventions match the domestic data provider: RSI (14), simple moving averages (20/50), Bollinger Bands (20,  $2\sigma$ ), MACD (12/26/9) and the KDJ (9/3/3) stochastic. These five are among the most widely used indicators in technical analysis and were chosen because together they cover the main families of signals rather than several variants of the same idea: momentum (RSI), trend (moving averages and MACD), volatility and mean reversion (Bollinger Bands), and stochastic oscillation (KDJ) [18]. Each indicator  $i$  is converted to a binary bullish score  $s_i \in \{0, 1\}$  by a small set of rules (Table 1), where  $s_i = 1$  means indicator  $i$  is currently bullish and  $s_i = 0$  means it is not. The rules favour *nascent* upward momentum (fresh crossovers, accelerating gaps, rebounds from oversold zones) over already-overextended readings, and each  $s_i$  is reported with its raw values and a short justification so both the aggregator and the user can see why a signal fired. The agent then outputs the total technical score  $S_{\text{tech}} = \sum_{i \in \mathcal{E}} s_i$ , that is, the number of enabled indicators that are currently bullish, together with the maximum attainable score  $S_{\text{max}} = |\mathcal{E}|$ , the number of enabled indicators, where  $\mathcal{E}$  is the user-enabled indicator set (all five by default). Reporting both lets the aggregator normalize the technical signal to  $[0, 1]$  however many indicators are active.

### 4.3 Fundamental Analysis Agent

The fundamental agent assesses the issuer’s financial health from its annual records. Unlike the technical agent its evidence window is horizon-independent: fundamentals change slowly, so it always retrieves the most recent *five fiscal years* for every risk profile. It reads per-ticker data (analyst summary, indicators, income statement, cash flow, balance sheet) and assembles five toggleable metric groups: valuation (P/E, P/B, EV/EBITDA, EPS)<sup>3</sup>, profitability (ROE, ROA, margins)<sup>4</sup>, growth (revenue and profit year over year), financial health

<sup>3</sup> <https://www.kotakneo.com/stockshaala/valuations/valuation-multiples-explained/>

<sup>4</sup> ROE: Return on Equity; ROA: Return on Assets.

**Table 1.** Bullish scoring rules. An indicator earns  $s_i = 1$  if any listed condition holds between the previous candle and the current one.

| Indicator Conditions for $s_i = 1$ (logical OR) |                                                                                                                                               |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| RSI                                             | crosses above 50; or $50 \leq \text{RSI} \leq 70$ and rising; or $\text{RSI} < 35$ and rising                                                 |
| MA                                              | golden cross (SMA20 crosses above SMA50); or $\text{SMA20} > \text{SMA50}$ with widening gap; or $\text{price} > \text{SMA20} > \text{SMA50}$ |
| BOLL                                            | price crosses above middle band; or above middle with increasing distance; or rebound off lower band                                          |
| MACD                                            | crosses above signal; or $\text{MACD} > \text{signal}$ with rising histogram; or histogram turns positive                                     |
| KDJ                                             | $K$ crosses above $D$ ; or $K > D$ with rising $J$ ; or $K < 20$ and rising                                                                   |

(current ratio, debt-to-equity, interest coverage) and cash flow (CFO, CAPEX, dividends)<sup>5</sup>, reporting the latest year as a snapshot with year-over-year growth and the multi-year trend. Ratios ill-defined for a sector (gross margin or debt-to-equity for banks) are annotated rather than reported misleadingly, and missing data (e.g. newly listed tickers) yields a “no data” marker instead of aborting a batch run. The resulting report is interpreted by the aggregator into a discrete `fundamental_health` label (strong/neutral/weak/N/A).

#### 4.4 Article Analysis Agent

The news agent captures information invisible to price and accounting data, in two stages. Offline, news articles on Vietnamese companies are collected and each is passed once through an LLM enrichment step that fills a `NewsExtraction` schema (the mentioned tickers, a sentiment label, and a short summary) and links it to stocks through an `Article_Stock` table, caching the sentiment and summary for reuse. Online, for a requested ticker the agent retrieves the linked articles within the orchestrator’s window and passes them to `GPT-4o`, which discards uninformative items and synthesizes a brief of the principal news, positive points and negatives/risks (or “no useful news”); this is the one place we allow mild generative freedom (temperature 0.2), yet it remains an analysis step. The source can be toggled off and empty windows handled gracefully, and the aggregator condenses the brief into an `article_sentiment` label (positive/neutral/negative/N/A).

#### 4.5 Aggregator Agent

The aggregator fuses the three streams into one recommendation. `GPT-4o` reads the agent outputs and emits a structured `InvestmentRecommendation` with a

<sup>5</sup> Cash Flow from Operations; Capital Expenditures.

summarization and a proposed verdict, entry, take-profit, stop-loss and holding period. Alongside the verdict it returns the analysis together with a score or a label for each component: a numeric technical score between 0 and 1, and qualitative labels for the other two streams. Fundamental health is one of **strong/neutral/weak** and news sentiment one of **positive/neutral/negative**, so unlike the technical gate they have no numeric cutoff, and only the clearly adverse label acts as a veto.

From the three signals, the technical score  $S_{\text{tech}}$ , a fundamental health label and a news sentiment label, each normalized to  $[0, 1]$ , the aggregator computes a confidence as their weighted mean. The weights are equal but are *renormalized over only the sources that carry data* (detected by Python flags, not trusted to the model), so a source with no data is dropped rather than dragging the score toward the midpoint.

The verdict is **Buy** only if the technical score reaches at least 60% of its attainable maximum (rounded up, so 3 of 5 when all indicators are enabled), the fundamental label is not **weak**, the news label is not **negative**, and the confidence clears a threshold of 0.55; otherwise **Wait**. For a **Buy** the entry, take-profit, stop-loss and holding period are clamped to hard, horizon-dependent bounds and must meet a minimum reward-to-risk ratio of 1.5, else the trade is downgraded to **Wait**. Moreover, any decoding failure resolves to a safe **Wait**.

#### 4.6 Agent Workflow

Having described what each agent computes, we now describe how they coordinate: the shared communication protocol, the types of interaction between agents, and the backbone LLMs. Figure 3 traces one analysis request through the pipeline.

**Shared state and communication protocol.** The agents do not call one another directly; they communicate through a single typed state object, **AgentState** (a **TypedDict**), that flows through the graph. It carries the execution mode, the user input, the risk profile, the target symbol, a **data\_selection** object (per-source and per-indicator toggles), the orchestrator’s plan, the collected **agent\_results**, and the final output. Every agent that consults the LLM does so through *structured decoding*: the model must return a valid instance of a Pydantic schema (the plan, the extracted signals, the recommendation) rather than free text, so each message exchanged in the graph is well-typed and machine-consumable rather than parsed heuristically. The **agent\_results** field is annotated with an **operator.or\_** reducer, letting the parallel analysis agents each write their partial result into the shared dictionary without overwriting one another.

**Types of agent interactions.** Three interaction patterns compose the pipeline (Fig. 3). First, *planning*: the orchestrator reads the request and risk profile and emits parameters for every downstream agent, so the analysis agents receive

a consistent time window and interval. Second, *running the analysis agents in parallel*: the technical, fundamental and article agents run at the same time and independently, and their results are combined once all of them finish, so no agent waits on another and a missing source does not stall the rest. Third, *veto-based aggregation*: the aggregator consumes the merged results and, rather than averaging opaque scores, applies the deterministic fusion and decision rules of Section 4.5, in which the technical score acts as the trigger while the fundamental and news signals act as vetoes. A source that is disabled or carries no data is detected by Python flags and dropped from the computation, so the interaction degrades gracefully as sources come and go.

**Backbone LLMs.** All agents share one backbone, OpenAI’s **GPT-4o**, accessed via structured decoding and chosen mainly for cost; since every agent talks to it through the same thin interface, it can be swapped for a more capable model through configuration without changing the pipeline. Temperature follows the “analyze versus decide” split: deterministic steps (orchestration, the aggregator’s signal extraction) run at 0, generative steps (news synthesis, narrative justification) at 0.2–0.3 – low but non-zero, since 0 yields stiff, near-duplicate text while higher values risk drifting from the evidence, so a small value keeps the prose varied yet grounded and runs largely reproducible. The aggregator caps its output tokens and retries once on truncation, and any unhandled error resolves to a safe `Wait`. Confining the model to language while the decisive computations stay in Python keeps the workflow reproducible and auditable.

## 5 Experiments

Backtesting is a central capability of Stockrium, serving its guiding purpose: letting users experiment with and verify a strategy on historical data before risking capital. We describe how the tool collects data, configures a strategy, runs and scores a backtest, and visualizes and compares results.

### 5.1 Data collection

Experiments require historical data, which the application collects and maintains itself rather than fetching live on demand. Daily and Intraday OHLCV<sup>6</sup> data for the tracked universe and the VN-Index (the benchmark) come from the SSI Fast Connect API<sup>7</sup> and are stored as a rolling multi-year history; company fundamentals and Vietnamese financial news are collected into the same store, the latter enriched offline into per-article sentiment (Section 4.4). Because the data is gathered ahead of time and standardized, every backtest runs over a consistent, reproducible snapshot. Our experiments use the VN30 constituents (30 symbols) over a three-year window, January 2023 to December 2025.

<sup>6</sup> Open, High, Low, Close, Volume

<sup>7</sup> Source: <https://guide.ssi.com.vn/ssi-products/tieng-viet/fastconnect-data>

## 5.2 Strategy configuration and comparison

A *strategy* in Stockrium is a configuration of the analysis: which agents are enabled, which technical indicators and fundamental metric groups feed them, and the risk horizon. The user sets this configuration, runs a backtest, and the tool records the run together with its metrics; multiple runs can be stored and compared side by side, so a user can see how, say, enabling the news agent, adding an indicator, or changing the horizon affects performance. Two baselines are available for reference: the unfiltered technical engine that drives the pipeline (which trades on every technical signal, without the LLM fusion and vetoes) and a passive buy-and-hold strategy.

## 5.3 Backtesting procedure

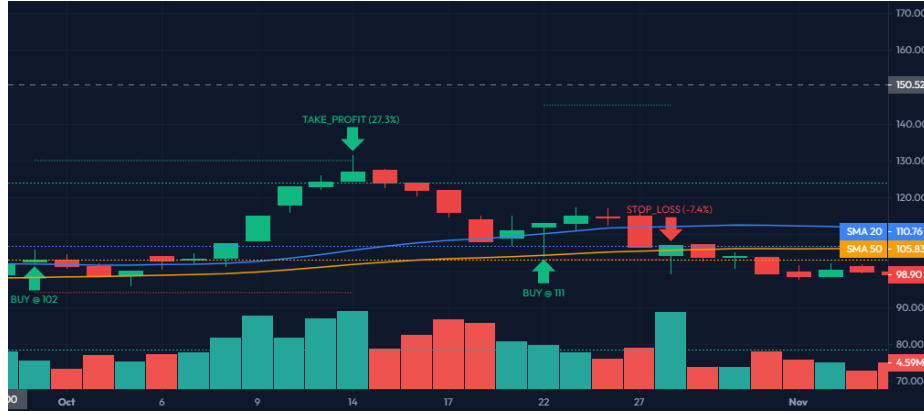
The backtest replays each trading record in order with no look-ahead: indicators at bar  $i$  use only data up to  $i$ , and a simulated entry fills at the open of bar  $i+1$ . To bound LLM cost, the full pipeline is invoked only on dates where the technical engine emits a signal; on those dates the orchestrator, the three analysis agents and the aggregator run exactly as in production (Section 4) to produce a buy/wait decision with entry, take-profit, stop-loss and maximum holding period. A Buy opens a position that is simulated forward candle by candle until the take-profit or stop-loss is hit or the holding limit expires, whichever comes first, and the realized return is recorded.

## 5.4 Performance metrics

Each trade yields a net return  $r = (p_{\text{exit}} - p_{\text{entry}})/p_{\text{entry}} - 2c$ , where  $c = 0.15\%$  is the per-side transaction cost. Over the  $n$  trades of a run we report the *win rate* (fraction with  $r > 0$ ); the *average return* per trade  $\bar{r}$ ; the compounded *total return*  $\prod_i (1 + r_i) - 1$  and its annualization  $(1 + \text{total})^{365/D} - 1$  over the  $D$ -day span; the *maximum drawdown*, the largest peak-to-trough drop of the equity curve  $\prod_i (1 + r_i)$ ; a per-trade *Sharpe ratio*  $\sqrt{n} \bar{r} / \sigma_r$ ; and the *profit factor*, gross profit over gross loss  $\sum_{r_i > 0} r_i / |\sum_{r_i < 0} r_i|$ . We additionally run significance tests: a one-sample  $t$ -test of returns against zero, a sign-permutation test, and the information coefficient between the pipeline’s confidence and the forward return. All figures are computed identically for the two baselines and pooled across market regimes and confidence tiers for the analysis in Section 6.

## 5.5 Trade visualization

For inspection, the tool overlays every simulated trade on the price chart, marking each entry and exit, so a user can see precisely where a strategy would have bought and sold and judge its behaviour visually before committing real capital (Fig. 4).



**Fig. 4.** Backtest visualization: simulated trades overlaid on the price chart with their entry and exit points marked.

## 6 Results and Analysis

We report results for the three-year evaluation window of Section 5. Aggregate figures are per-symbol means, with the median given where the distribution is skewed by outliers.

### 6.1 Aggregate performance

Table 2 reports the three-year run for the full pipeline, the unfiltered technical engine (all five indicators), each single indicator run in isolation, and passive buy-and-hold. The pipeline’s most consistent effect is *selectivity*: requiring a clear technical trigger and vetoing on weak fundamentals or negative news, it takes far fewer positions than the engine (450 vs. 759 trades) and cuts average maximum drawdown by about a third ( $-17.7\%$  vs.  $-26.8\%$ ), while keeping median total return comparable ( $23.2\%$  vs.  $21.3\%$ ) at a similar Sharpe ratio. The per-indicator rows show that no single indicator dominates: KDJ and Bollinger are strongest in isolation (Sharpe 0.72 and 0.69, median return near 28%), MACD and MA are middling, and RSI is weakest (0.50, 7.5%); all of them, like the combined engine, incur large drawdowns of 17–28%, because a single-indicator strategy enters on a signal and holds to a fixed horizon without a stop-loss (MA excepted, with the fewest trades and the smallest drawdown,  $-17.0\%$ ). Combining all five thus does not beat the best single indicator on return and keeps the large drawdown, whereas the pipeline’s fundamental and news vetoes and risk-managed exits cut drawdown to  $-17.7\%$  and give the highest average return per trade (2.4%). The wide spread across indicators also supports the configurable design: a user might well prefer KDJ or a low-drawdown MA setup to the full five.

**Table 2.** Backtest performance over the VN30 universe (three-year window, per-symbol mean; total return is the median)

| Strategy             | Trades | Win   | Avg/trd | Tot. ret. | Sharpe | Max DD |
|----------------------|--------|-------|---------|-----------|--------|--------|
| RSI                  | 824    | 50.2% | 1.0%    | 7.5%      | 0.50   | −27.6% |
| MA                   | 407    | 51.6% | 0.9%    | 11.8%     | 0.58   | −17.0% |
| Bollinger            | 806    | 50.7% | 1.6%    | 28.5%     | 0.69   | −27.0% |
| MACD                 | 702    | 51.3% | 1.3%    | 11.7%     | 0.60   | −24.7% |
| KDJ                  | 890    | 52.1% | 1.3%    | 28.1%     | 0.72   | −26.3% |
| Technical (all five) | 759    | 52.0% | 1.6%    | 21.3%     | 0.70   | −26.8% |
| Pipeline             | 450    | 48.6% | 2.4%    | 23.2%     | 0.67   | −17.7% |
| Buy-and-hold         | n/a    | n/a   | n/a     | 85.5%     | n/a    | n/a    |

## 6.2 Benchmark comparison and regimes

Against the unfiltered engine the pipeline reduced drawdown substantially while improving total return on 13 of 30 symbols: roughly a wash on raw return but a clear gain on risk. Against buy-and-hold it is less favourable, with only 4 of 30 symbols beating it (median buy-and-hold 85.5%), as expected for a selective, partially invested, long-only strategy over a bullish 2023–2025 recovery, whose value lies instead in sitting out pullbacks (hence the smaller drawdown); against random entries of equal count it sat around the 63% percentile. By regime, it opened almost no positions in downtrends (one trade of 450), concentrating on sideways (385 trades, 53.2% win, 2.1% average) and uptrending (64, 35.9%, 1.8%) markets, a direct consequence of the buy/wait logic.

## 6.3 Calibration and statistical significance

The three-year run lets us probe the confidence score and significance. High-confidence trades won marginally more often than medium ones (50.9% vs. 49.3% pooled; 48.4% vs. 43.7% per symbol), a weak positive ordering, with no low-tier trades by design (low confidence maps to *Wait*); a walk-forward over  $\sim 20$  rolling windows per symbol was net-positive in about 20% of windows. On significance, with an average of about 15 trades per symbol a one-sample  $t$ -test against zero and a sign-permutation test each reach  $p < 0.05$  for only 2 of 30 symbols, and the mean confidence–return information coefficient is weakly positive (0.035) though not significant in aggregate. We thus read the results as evidence the pipeline behaves as designed (selective, drawdown-aware, downtrend-avoiding) rather than as a statistically established return edge; because the window is predominantly bullish with few in-position declines, establishing such an edge would need a larger universe and windows spanning full bear markets.

## 6.4 Discussion

The results are best read not as beating the market but as evidence the system meets its goals of explainability and risk control: more selective than the un-

filtered engine, it cuts drawdown by about a third and matches its three-year return at comparable Sharpe with less risk, all traceable to explicit rules under the hybrid design. They are also an early trial of a backtesting tool we built ourselves rather than a definitive verdict, since any result depends on the user’s own choices of indicators and models; what we mainly show is the ability to run backtests flexibly so a strategy can be judged in quantitative rather than purely qualitative terms.

Several caveats apply: the window is predominantly bullish with few trades per symbol, so significance tests are underpowered; the VN30 universe uses current membership (survivorship bias); the long-only strategy structurally lags buy-and-hold in rallies; returns include transaction costs but not slippage or portfolio-level allocation; and the pipeline depends on a proprietary LLM whose narratives may still err. The system is decision-support only, not financial advice, and confined to Vietnamese equities, though its auditable decision layer lets a user or regulator inspect any recommendation.

## 7 Conclusion

We presented an agentic decision-support system for Vietnamese equities that unifies technical, fundamental and news-sentiment analysis with recommendation generation. Its core is a multi-agent LangGraph pipeline (an orchestrator, three parallel analysis agents and an aggregator) governed by a single principle: the language model analyzes, while deterministic code decides, yielding explainable, reproducible buy/wait recommendations with a renormalized confidence score and horizon-dependent risk targets. Demo videos of the mobile and web applications are available online.<sup>8</sup>

Backtesting on VN30 over a three-year window shows the system behaves as designed: markedly more selective than the unfiltered engine, cutting average drawdown by about a third and almost never trading into downtrends, while matching its median return at a comparable Sharpe ratio with much lower risk. As the window is predominantly bullish with few trades, we claim not a statistically established return edge but disciplined, risk-aware, transparent behaviour. Within its scope (analysis only, no execution, Vietnamese equities alone), it serves not just amateurs testing suggestions but semi-professional users probing the fundamentals of individual stocks or whole baskets, and researchers who gain a ready testbed into which new AI trading algorithms can be integrated and evaluated. Future work targets longer, multi-regime evaluation over broader universes with point-in-time membership, portfolio-level backtesting with slippage and exposure-adjusted metrics, and better confidence calibration.

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<sup>8</sup> [https://drive.google.com/drive/folders/1FN3j\\_Un-9gzblFytVNeGLWqDyC-ZJYn8](https://drive.google.com/drive/folders/1FN3j_Un-9gzblFytVNeGLWqDyC-ZJYn8)

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